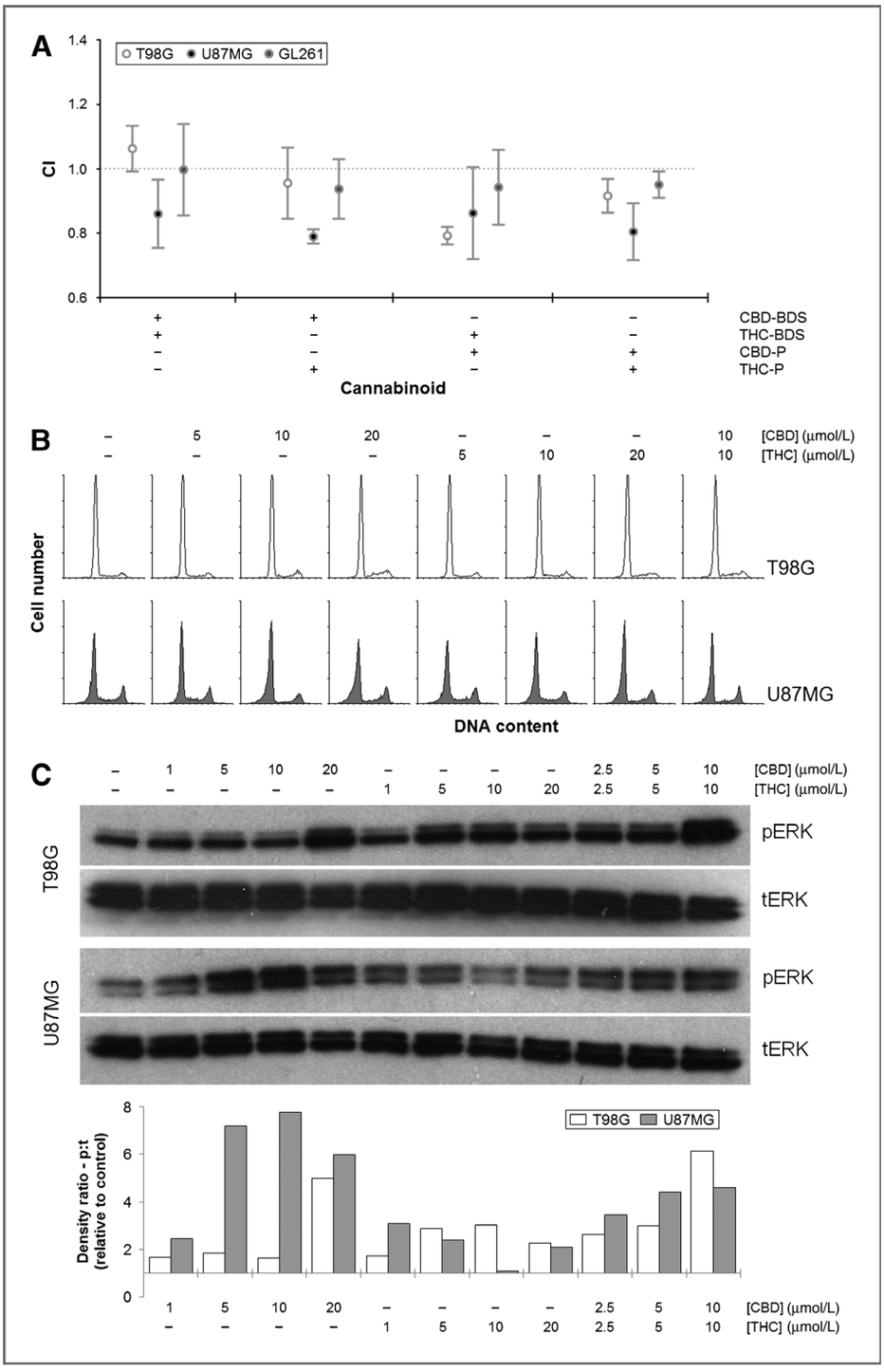

Figure 2. Effect of combining CBD with THC on cell number and cell signaling molecules. T98G, U87MG, and GL261 were cultured with both forms of each cannabinoid concomitantly, and the nonexclusive CI for the 50% unaffected fraction was assessed by median effect modeling. CI value of >1 indicates antagonism; CI of 1 indicates additivity, and CI < 1 synergy. Most combinations resulted in outcomes that were hyperadditive (A). Cell-cycle profiles determined by propidium iodide staining in T98G and U87MG are shown in B. Only the pure forms of THC and CBD were studied. The figure also details the effect of combining equimolar amounts of each cannabinoid. T98G and U87MG cells were also cultured with CBD and THC in their P form either individually or together at the concentrations indicated for 4 hours before Western blotting for pERK and tERK (C). The levels of pERK appeared to be the protein changed most often. Each data point/blot represents the mean and SDs of at least three separate experiments.

expression of key protagonists in signaling cascades that we have previously identified as being involved in the function of cannabinoids (3, 4, 11). Results showed that the greatest change was seen in $MAPK^{ERK1/2}$, where culturing with either CBD-P or THC-P for 4 hours altered its phosphorylation (Fig. 2C). Specifically, dose-dependent increases in the ratio of phosphorylated to total ERK was seen when T98G was cultured with either CBD or THC. A similar increase in levels was seen in U87MG cultured with CBD; however, they were reduced when treated with THC. The effect of AKT expression was less clear, nevertheless, there was a hint that cannabinoids reduced the expression/phosphorylation (data not shown).

Downloaded from http://aacrjournals.org/mct/article-pdf/13/12/2955/2328449/2955.pdf by guest on 15 March 2025